

Protective effects of green tea extracts (polyphenon E and EGCG) on human cervical lesions.

We investigated clinical efficacy of green tea extracts (polyphenon E; poly E and (-)-epigallocatechin-3-gallate [EGCG]) delivered in a form of ointment or capsule in patients with human papilloma virus (HPV) infected cervical lesions. Fifty-one patients with cervical lesions (chronic cervicitis, mild dysplasia, moderate dysplasia and severe dysplasia) were divided into four groups, as compared with 39 untreated patients as a control. Poly E ointment was applied locally to 27 patients twice a week. For oral delivery, a 200 mg of poly E or EGCG capsule was taken orally every day for eight to 12 weeks. In the study, 20 out of 27 patients (74%) under poly E ointment therapy showed a response. Six out of eight patients under poly E ointment plus poly E capsule therapy (75%) showed a response, and three out of six patients (50%) under poly E capsule therapy showed a response. Six out of 10 patients (60%) under EGCG capsule therapy showed a response. Overall, a 69% response rate (35/51) was noted for treatment with green tea extracts, as compared with a 10% response rate (4/39) in untreated controls ($P < 0.05$). Thus, the data collected here demonstrated that green tea extracts in a form of ointment and capsule are effective for treating cervical lesions, suggesting that green tea extracts can be a potential therapy regimen for patients with HPV infected cervical lesions.